A finite-difference measure converse and a Jacobi-symbol determinant counterexample

Chasse–Csordas Conjecture 2.22 and Sun Conjecture 5.6(ii)

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Abstract

We give complete proofs concerning two previously stated conjectures. First, if a real polynomial p of degree $n \geq 2$ satisfies

$$\operatorname{meas}\left\{x \in \mathbb{R} : \frac{p(x+1)-p(x)}{p(x)} \geq \lambda\right\} = \frac{n}{\lambda} \qquad (\lambda > 0),$$

then all its zeros are real and simple, and consecutive zeros are at least one unit apart. This proves the implication in Chasse–Csordas Conjecture 2.22. In fact, the identity on any nonempty open interval of positive levels suffices. A rational measure-rigidity argument gives positive simple real poles after cancellation; a divisor-count identity then recovers the roots and multiplicities of the original polynomial. The inverse-branch mechanism is an adaptation of Szegő's 1934 method, and is not claimed as a historically new principle.

Second, for Sun's original truncated Jacobi-symbol determinant $D_n = \{5, 5\}_n$, we prove

$$D_{5469} \equiv 2 \pmod 3, \qquad D_{5469} \equiv 1089 \pmod{1823}, \qquad \left(\frac{D_{5469}}{5469}\right) = -1.$$

Since $5469 \equiv 9 \pmod{60}$ and $5469 > 69$, this is a counterexample to Conjecture 5.6(ii). The proof uses a complementary inverse minor, character diagonalization, and exact polynomial arithmetic. A complete standard-library Python certificate is embedded in this single source. Mathematical conclusions, historical attribution, and unverified priority claims are kept separate.

1 The finite-difference measure conjecture

For a real polynomial p of degree $n \geq 2$, define

$$F_p(x) = \frac{p(x+1)-p(x)}{p(x)}.$$

Its domain excludes the real zeros of p. Removing or filling any removable singularities changes a level set only at finitely many points, so it does not change its Lebesgue measure. For a polynomial with real zeros $r_1 \leq \cdots \leq r_n$, listed with multiplicities, write

$$\mu(p) = \min_{1 \leq j < n} (r_{j+1} - r_j).$$

Thus $\mu(p) \geq 1$ includes simplicity of every zero.

Chasse and Csordas [1, Conjecture 2.22, p. 88] ask whether the exact identity

$$\operatorname{meas}\{x \in \mathbb{R} : F_p(x) \geq \lambda\} = \frac{n}{\lambda} \quad \text{for every } \lambda > 0 \tag{1}$$

forces p to have only real zeros and $\mu(p) \geq 1$. No real-rootedness or simplicity hypothesis is imposed in this converse. We prove it in the following slightly stronger form.

Theorem 1.1 (Finite-difference measure characterization)**.** *Let $p \in \mathbb{R}[x]$ have degree $n \geq 2$. The following are equivalent.*

1. *All zeros of p are real and $\mu(p) \geq 1$.*
2. *Equation (1) holds for every $\lambda > 0$.*
3. *There is a nonempty open interval $I \subset (0, \infty)$ on which (1) holds for every $\lambda \in I$.*

In particular, the implication in Chasse–Csordas Conjecture 2.22 holds under its original hypotheses.

The forward implication is included for completeness and is not a new claim: it is Theorem 88 of Chasse's 2011 dissertation [2, pp. 45–46]; the converse appears there as Conjecture 89. The main algebraic mechanism below has a substantial precedent in Szegő's solution of Pólya's Problem 107 [3, pp. 19–20]. We provide all details needed for the present local level law and for the recovery of the original polynomial after cancellation.

2 A local rational measure-rigidity theorem

Theorem 2.1 (Local Boole-law rigidity)**.** *Let $F = A/B$ be a reduced nonconstant real rational function such that*

$$F(x) = \frac{c}{x} + O(x^{-2}) \quad (x \longrightarrow \infty), \qquad c > 0. \tag{2}$$

If, on some nonempty open interval $I \subset (0, \infty)$,

$$\operatorname{meas}\{x \in \mathbb{R} : F(x) \geq \lambda\} = \frac{c}{\lambda} \qquad (\lambda \in I), \tag{3}$$

then

$$F(x) = \sum_{j=1}^{m} \frac{\rho_j}{x - b_j}, \qquad b_1 < \cdots < b_m \in \mathbb{R}, \quad \rho_j > 0, \quad \sum_{j=1}^{m} \rho_j = c. \tag{4}$$

In particular, every pole of the reduced rational function is simple and real. Here $m = \deg B$.

Proof. Normalize B to be monic. Equation (2) implies $m = \deg B \geq 1$, $\deg A = m - 1$, and that the leading coefficient of A is c.

Step 1: an identity among inverse branches. The polynomial

$$L(X, T) = TB(X) - A(X)$$

has only finitely many exceptional nonzero parameter values at which its roots are not simple. Indeed, it is separable over $\mathbb{C}(T)$, as also follows from the irreducibility argument in Step 2. Choose a connected open subinterval $J \subset I$ avoiding these values. Its m roots can be labeled by analytic algebraic branches $\alpha_1(T), \ldots, \alpha_m(T)$ on a complex neighborhood of a point of J and then continued along J. No root is a pole of F, since A and B are coprime. Their real or nonreal status, and the crossing direction of every real root, are fixed on J.

For $T > 0$, the set $\{x : F(x) \geq T\}$ is bounded because $F(x)$ tends to zero at both ends of the real axis. Its endpoints are real roots of $F(x) = T$ and real poles of F. The poles contribute a constant C. A downward crossing contributes its coordinate positively to the length, an upward crossing negatively, and a nonreal root does not contribute. Hence the assumed identity is

$$C + \sum_{i=1}^{m} e_i \alpha_i(T) = \frac{c}{T}, \qquad e_i \in \{-1, 0, 1\}, \tag{5}$$

where the e_i are constant on J. This analytic identity is an identity of algebraic germs: its difference vanishes on a real interval and therefore on the complex neighborhood by the identity theorem. We may consequently apply automorphisms of a splitting field of L over $\mathbb{C}(T)$.

Step 2: transitivity excludes missing branches and wrong signs. The polynomial $TB(X) - A(X)$ is irreducible in $\mathbb{C}[X, T]$. In any factorization, its degree one in T forces one factor to belong to $\mathbb{C}[X]$; that factor would divide both A and B, and is therefore constant. It is primitive as a polynomial in X over $\mathbb{C}[T]$: a common nonconstant parameter factor would give $A = t_0 B$ for some t_0, contrary to their different degrees. Gauss's lemma gives irreducibility over $\mathbb{C}(T)$, and characteristic zero gives separability. Thus its splitting-field Galois group G acts transitively on the m roots.

Let $s = \sum_i e_i$ and let S_B be the sum of the roots of B, with multiplicities. Average all G-conjugates of (5). Each root occurs with average coefficient s/m, so

$$C + \frac{s}{m} \sum_{i=1}^{m} \alpha_i(T) = \frac{c}{T}.$$

By Vieta's formula,

$$\sum_{i=1}^{m} \alpha_i(T) = S_B + \frac{c}{T}.$$

Comparison of the coefficients of $1/T$ gives $s = m$. Since every $e_i \le 1$, this forces

$$e_1 = \cdots = e_m = 1.$$

Every level root on J is therefore real and is a downward crossing.

Step 3: the poles are simple, real, and positive. Fix $T \in J$ and order the level roots $\alpha_1 < \cdots < \alpha_m$. At minus infinity $F - T < 0$, whereas immediately to the left of each α_i it is positive and immediately to the right it is negative. Each of the m disjoint intervals

$$(-\infty, \alpha_1), \quad (\alpha_1, \alpha_2), \quad \ldots, \quad (\alpha_{m-1}, \alpha_m)$$

must contain a pole. Otherwise continuity would require an additional level root, or an upward crossing, to change the sign from negative to positive. Since B has degree m, these exhaust all its zeros: there is exactly one simple real pole in each interval and no others. Across each such pole the sign changes from negative on the left to positive on the right. The residue is therefore positive. The rational function is proper, so its partial-fraction expansion is (4); comparison at infinity gives $\sum_j \rho_j = c$. □

Remark 2.2 (Attribution of the mechanism). Szegő's 1934 converse for rational Lebesgue-measure preservers already uses signed algebraic inverse branches, Vieta's formula, irreducibility, and transitive monodromy to exclude a missing branch or an incorrect sign [3, pp. 19–20]. The Galois average above is an adaptation of that mechanism to a local positive-tail identity with a prescribed coefficient at infinity. We do not present the underlying principle as new, or infer worldwide priority for the finite-difference application from a bounded search.

3 Recovering all polynomial zeros after cancellation

Proof of Theorem 1.1. The implication (2)⇒(3) is immediate. We first prove (3)⇒(1).

For any real polynomial of degree n,

$$F_p(x) = \frac{p(x+1) - p(x)}{p(x)} = \frac{n}{x} + O(x^{-2}).$$

Reduce this rational function before applying Theorem 2.1. The denominator degree m after reduction may be smaller than n, but the coefficient at infinity remains n. We obtain

$$R(x) := \frac{p(x+1)}{p(x)} = 1 + F_p(x) = 1 + \sum_{j=1}^{m} \frac{\rho_j}{x - b_j}, \qquad \rho_j > 0.$$

No nonreal root can be lost through all cancellations. Suppose that p has a nonreal root. Among the roots on one of the corresponding horizontal lines, choose z with maximal real part. Then $p(z+1) \neq 0$, so R has a nonreal pole at z. This contradicts the real-pole conclusion of Theorem 2.1. Hence all roots of p are real.

The reduced quotient has strictly alternating zeros and poles. For real x away from the poles,

$$R'(x) = -\sum_{j=1}^{m} \frac{\rho_j}{(x-b_j)^2} < 0.$$

Its limit at both infinities is 1, and its limits at each pole are $-\infty$ from the left and $+\infty$ from the right. It follows that R has exactly one simple zero to the left of b_1 and one in each interval (b_j, b_{j+1}), with none to the right of b_m. These m zeros exhaust its numerator degree. Denoting them by $a_1, \ldots, a_m$, their order is

$$a_1 < b_1 < a_2 < b_2 < \cdots < a_m < b_m. \tag{6}$$

A divisor count recovers multiplicity and spacing. Let $N_p(t)$ count the zeros of p in $(-\infty, t]$, with their original multiplicities. The zeros of $p(x+1)$ are the zeros of p shifted left by one. Common factors cancel equally in the numerator and denominator, so for every real t,

$$N_p(t+1) - N_p(t) = \#\{j : a_j \leq t\} - \#\{j : b_j \leq t\} \in \{0, 1\}. \tag{7}$$

The final inclusion follows from (6). If p had a multiple root, or two distinct roots less than one apart, we could place at least two roots, counted with multiplicities, strictly inside some interval $(t, t+1)$. The left side of (7) would then be at least two, a contradiction. All roots are consequently simple and $\mu(p) \geq 1$.

The known forward implication. Suppose now that $r_1 < \cdots < r_n$ are the roots of p and $r_{j+1} - r_j \geq 1$. Partial fractions give

$$F_p(x) = \sum_{j=1}^{n} \frac{\rho_j}{x - r_j}, \qquad \rho_j = \frac{p(r_j+1)}{p'(r_j)} = \prod_{k \neq j} \frac{r_j + 1 - r_k}{r_j - r_k} \geq 0.$$

Every factor is nonnegative by the spacing assumption. Zero residues represent removable poles; discard them, retaining poles $b_1 < \cdots < b_m$ with positive residues. Their sum is n by the coefficient at infinity. For each $\lambda > 0$, strict decrease between poles shows that the superlevel set is, up to finitely many endpoints,

$$\bigcup_{j=1}^{m} (b_j, \alpha_j],$$

where α_j is the unique level root to the right of b_j and before the next pole, or on (b_m, ∞) for $j = m$. Vieta's formula for $\lambda B - A$ gives

$$\operatorname{meas}\{F_p \geq \lambda\} = \sum_j (\alpha_j - b_j) = \frac{n}{\lambda}.$$

This proves (1)⇒(2), and finishes the proof. □

Remark 3.1 (Why reduction cannot be skipped). For $p(x) = \prod_{j=0}^{n-1}(x - a - j)$,

$$F_p(x) = \frac{n}{x - a - (n-1)}.$$

The reduced denominator has degree one, although $\deg p = n$. This example attains $\mu(p) = 1$ and obeys the exact measure law. Conversely, for $p(x) = x^2(x-1)$ one obtains

$$F_p(x) = -\frac{1}{x} + \frac{4}{x-1}, \qquad \operatorname{meas}\{F_p \geq 1\} = 2\sqrt{5} - 1 \neq 3.$$

The negative residue records the failure of the required root spacing. These examples illustrate why the proof must retain the original divisor multiplicities after reducing the quotient.

4 The unchanged Sun conjecture and its counterexample

Sun defines, for an odd integer $n > 1$,

$$\{c, d\}_n = \det\left[\left(\frac{j^2 + cjk + dk^2}{n}\right)\right]_{1<j,k<n-1}, \tag{8}$$

where the entries are Jacobi symbols. The indices are all integers $2, \ldots, n-2$, without a coprimality restriction. Conjecture 5.6(ii) in [4, p. 23] asserts that $\left(\frac{\{5,5\}_n}{n}\right) = 0$ whenever $n \equiv 11, 19 \pmod{20}$, or whenever $n \equiv 9 \pmod{60}$ and $n > 69$. The assumption that n is a sum of two squares belongs to part (i) and is absent from part (ii).

Li and Yan [5, Theorem 1.1(f), Remark 1.2] prove the $11, 19 \pmod{20}$ cases of part (ii). Their stated results leave the $9 \pmod{60}$ clause untreated. The counterexample below addresses exactly that clause.

Theorem 4.1. *Set $p = 1823$, $n = 3p = 5469$, and $D_n = \{5, 5\}_n$, with the definition (8). Then*

$$D_n \equiv 2 \pmod{3}, \qquad D_n \equiv 1089 \pmod{p}. \tag{9}$$

Equivalently, $D_n \equiv 2912 \pmod{5469}$. Consequently $\gcd(D_n, n) = 1$ and

$$\left(\frac{D_n}{n}\right) = -1.$$

In particular, Sun's Conjecture 5.6(ii), as stated in [4], is false.

The modulus $n = 5469$ is odd, exceeds 69, and is congruent to 9 modulo 60. Since $1089 = 33^2$ is nonzero modulo 1823, (9) gives

$$\left(\frac{D_n}{3p}\right) = \left(\frac{2}{3}\right)\left(\frac{33^2}{1823}\right) = (-1)(1) = -1.$$

Thus proving the two displayed congruences suffices. No exact decimal expansion of the large integer D_n is needed.

What is being proved computationally. This is a finite, exact certificate for a counterexample to a universal assertion. It is not an extrapolation from a finite range of successful tests. The matrices, fields, loop bounds, polynomial identities and final residues are all specified below. The certificate uses integer arithmetic throughout.

5 Restoring the deleted indices

Write $q(X) = X^2 + 5X + 5$. For an odd prime r, define the full matrix

$$A_r = \left[\left(\frac{j^2 + 5jk + 5k^2}{r}\right)\right]_{j,k \in \mathbb{F}_r}.$$

The row and column orders are the same. Define A_n by the same full matrix with Jacobi entries when n is composite. If $n = 3p$, with $p \neq 3$, the Chinese remainder bijection $\mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{F}_3 \times \mathbb{F}_p$ gives

$$A_n \sim A_3 \otimes A_p. \tag{10}$$

Here $\sim$ is simultaneous permutation of rows and columns, so it preserves determinants. Equality of entries follows from $\left(\frac{a}{3p}\right) = \left(\frac{a}{3}\right)\left(\frac{a}{p}\right)$. The determinant D_n is obtained from A_n by deleting the three rows and columns indexed by

$$\mathcal{E} = \{0, 1, -1\}.$$

Under the Chinese remainder ordering these indices become $(0,0), (1,1), (-1,-1)$, respectively.

Lemma 5.1 (Complementary principal minor)**.** *If M is an invertible matrix over a field and I is a set of indices, then*

$$\det M_{I^c,I^c} = \det M \ \det(M^{-1})_{I,I}.$$

Proof. After a simultaneous permutation, partition M into blocks indexed by I^c, I. If its upper-left block is invertible, block elimination and the Schur complement prove the formula. Multiplying by suitable powers of $\det M$ makes the identity polynomial in the matrix entries. The generic identity therefore holds for every invertible M, including cases where the upper-left block is singular. □

A direct calculation gives

$$A_3 = \begin{pmatrix} 0 & -1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}, \qquad \det A_3 = -4, \qquad A_3^{-1} = \begin{pmatrix} 0 & 1/2 & 1/2 \\ -1/2 & -1/4 & 1/4 \\ -1/2 & 1/4 & -1/4 \end{pmatrix}. \tag{11}$$

These formulas are valid over each field used below.

Proposition 5.2 (Three-coordinate reduction)**.** *Let K be a field of odd characteristic, let $p \neq 3$ be an odd prime, and suppose A_p is invertible over K. If the entries of $B = A_p^{-1}$ at $0, 1, -1$ have the form*

$$B_{\mathcal{E},\mathcal{E}} = \begin{pmatrix} 0 & u & u \\ -u & a & b \\ -u & b & a \end{pmatrix}, \tag{12}$$

then, as an identity in K,

$$D_{3p} = (-4)^p (\det A_p)^3 \, \frac{u^2(a+b)}{8}. \tag{13}$$

Proof. By (10), $\det A_{3p} = (-4)^p (\det A_p)^3$ and $A_{3p}^{-1} = A_3^{-1} \otimes B$ in the corresponding order. Its principal submatrix at the three deleted indices is

$$C = \begin{pmatrix} 0 & u/2 & u/2 \\ u/2 & -a/4 & b/4 \\ u/2 & b/4 & -a/4 \end{pmatrix}.$$

Expansion gives $\det C = u^2(a+b)/8$. Lemma 5.1 proves (13). □

Formula (13) shows the necessary additional condition: invertibility of the full local matrices alone does not ensure nonvanishing after truncation. The inverse entries must also satisfy $a+b \neq 0$. This is the cancellation that the certificate will check. Restoration, character diagonalization and Chinese remainder decomposition are established tools; see in particular [5, §1.2, Lemmas 2.1 and 2.4]. The complementary inverse minor is used here to prove nonvanishing for the original truncated determinant.

6 The congruence modulo 1823

We first work over $\mathbb{F}_p$, for a prime $p > 3$ with $p \equiv 3 \pmod{20}$. Put $h = (p-1)/2$, and define

$$q(X)^h = \sum_{s=0}^{p-1} a_s X^s \quad \text{in } \mathbb{F}_p[X]. \tag{14}$$

Quadratic reciprocity gives $\left(\frac{5}{p}\right) = -1$, so $a_0 = -1$ and $a_{p-1} = 1$.

Lemma 6.1. *Suppose $a_s \neq 0$ for $1 \leq s \leq p-2$. Define*

$$P = \prod_{s=1}^{p-2} a_s, \qquad E = \sum_{\substack{1 \leq s \leq p-2 \\ s \text{ even}}} a_s^{-1}.$$

Then

$$\det A_p = P, \qquad D_{3p} = -P^3 E \quad \textit{in } \mathbb{F}_p. \tag{15}$$

Proof. Let e_0 be the vector supported at zero, and let v_0 be zero at zero and one at every nonzero element of $\mathbb{F}_p$. For $1 \leq s \leq p-2$, let $v_s(0) = 0$ and $v_s(j) = j^s$ for $j \neq 0$. The vectors $e_0, v_0, v_1, \ldots, v_{p-2}$ form a basis. For nonzero j, k, Euler's criterion gives

$$(A_p)_{j,k} = q(j/k)^h = \sum_{t=0}^{p-1} a_t j^t k^{-t}.$$

Using $\sum_{k \neq 0} k^r = -1$ when $p-1 \mid r$ and zero otherwise, we obtain

$$A_p v_s = -a_s v_s \quad (1 \leq s \leq p-2), \qquad A_p e_0 = v_0, \qquad A_p v_0 = e_0.$$

The last equality uses $a_0 + a_{p-1} = 0$ and $-(p-1) = 1$ in $\mathbb{F}_p$. The two-dimensional block on e_0, v_0 has determinant -1. There are $p-2$ remaining, hence an odd number of, eigenvalues. It follows that

$$\det A_p = -\prod_{s=1}^{p-2} (-a_s) = P.$$

For a nonzero index k, the coordinate vector satisfies

$$e_k = -v_0 - \sum_{s=1}^{p-2} k^{-s} v_s.$$

Applying A_p^{-1} gives

$$A_p^{-1} e_k = -e_0 + \sum_{s=1}^{p-2} a_s^{-1} k^{-s} v_s.$$

Also $A_p^{-1} e_0 = v_0$. Thus (12) holds with

$$u = -1, \quad a = \sum_{s=1}^{p-2} a_s^{-1}, \quad b = \sum_{s=1}^{p-2} (-1)^s a_s^{-1}, \qquad a + b = 2E.$$

Proposition 5.2 and $(-4)^p = -4$ in $\mathbb{F}_p$ now give $D_{3p} = (-4)P^3(2E)/8 = -P^3 E$. □

6.1 Exact specialization

The complete calculation in Appendix A gives, for $p = 1823$,

$$\begin{gathered} a_s \neq 0 \quad (1 \leq s \leq 1821), \\ P = 697, \qquad E = 1662, \qquad a = 921, \quad b = 580. \end{gathered} \tag{16}$$

In particular, $a + b = 1501 = 2E$ in $\mathbb{F}_{1823}$, and the complementary inverse determinant is

$$\frac{E}{4} = 1327.$$

The full determinant is $(-4)^{1823}697^3 = 641$ in this field. Therefore

$$D_{5469} = 641 \cdot 1327 = -697^3 \cdot 1662 = 1089 \quad \text{in } \mathbb{F}_{1823}.$$

The program constructs every coefficient in (14) by 911 multiplications by $X^2 + 5X + 5$. It checks every required nonzero coefficient before taking any inverse. The value $E = 1662 \neq 0$, in addition to $P \neq 0$, is the operative noncancellation certificate.

7 The congruence modulo 3

The prime local matrix A_p can also be studied over a field whose characteristic differs from p. We give all the details needed for $p = 1823$ and $K = \mathbb{F}_3$. Set $m = p - 1 = 1822$. The number 5 generates $\mathbb{F}_p^\times$. Indeed, $m = 2 \cdot 911$, both p and 911 are prime, and the program verifies $5^{911} \not\equiv 1 \pmod{p}$ and $5^2 \not\equiv 1 \pmod{p}$.

Order the nonzero indices as $1, 5, \ldots, 5^{m-1}$, and define

$$f_k = \left(\frac{q(5^k)}{p}\right), \qquad F(X) = \sum_{k=0}^{m-1} f_k X^k, \qquad S(X) = 1 + X + \cdots + X^{m-1}, \tag{17}$$

with the coefficients of F reduced modulo 3. The nonzero-index block of A_p is the circulant $(f_{i-j})_{0 \leq i,j < m}$, where subscripts are taken modulo m.

Its row sum is zero already over the integers. To see this, the discriminant of q is $5 \neq 0 \pmod{p}$, and $\sum_{x \in \mathbb{F}_p} \left(\frac{q(x)}{p}\right) = -1$. The latter identity follows by completing the square and counting solutions of $y^2 = x^2 - d$, $d \neq 0$: the bijection $(x - y, x + y)$ identifies these solutions with pairs of nonzero elements with product d, of which there are $p - 1$. Removing $x = 0$, whose symbol is -1, gives a zero row sum. In particular $F(1) = 0$.

Since $3 \nmid m$, we have the direct sum of the constant vectors and the subspace of nonzero-index vectors with coordinate sum zero. On the two-dimensional space generated by e_0, v_0, the full matrix has the block

$$\begin{pmatrix} 0 & -m \\ 1 & 0 \end{pmatrix},$$

whose determinant is m. On the complementary space, the circulant has eigenvalues $F(\zeta)$, where ζ runs over the nontrivial m-th roots of unity in an algebraic closure of $\mathbb{F}_3$. The inversion of ζ merely permutes the eigenvalues. Consequently

$$\det A_p = m \operatorname{Res}(S, F). \tag{18}$$

Assume F is invertible modulo S. Compute H_0, of degree at most $m - 2$, such that $FH_0 \equiv 1 \pmod{S}$, and set

$$H = H_0 - \frac{H_0(1)}{m} S. \tag{19}$$

Then $H(1) = 0$, and

$$FH \equiv 1 - \frac{1}{m}S \pmod{X^m - 1}.$$

This follows by checking the congruence modulo the coprime polynomials S and $X - 1$. Thus the circulant defined by the coefficients of H is the inverse on the sum-zero subspace and vanishes on the constants. If

$$u = m^{-1}, \qquad a = [X^0]H, \qquad b = [X^{m/2}]H,$$

the entries of A_p^{-1} have exactly the form (12). The entries involving zero follow from the inverse of the displayed two-dimensional block. The equalities at $1, -1$ use $5^{m/2} = -1$.

The polynomial Euclidean algorithm and resultant computation in Appendix A give the exact data

$$\operatorname{Res}(S, F) = 2, \qquad u = 1, \qquad a = 1, \qquad b = 0 \quad \text{in } \mathbb{F}_3. \tag{20}$$

The inverse returned by the Euclidean algorithm is verified by a separate multiplication and polynomial remainder calculation. Equations (18) and (20) give $\det A_p = 2$, while

$$\det C = \frac{1^2(1+0)}{8} = 2 \quad \text{in } \mathbb{F}_3.$$

Finally, Proposition 5.2 yields

$$D_{5469} = (-4)^{1823} \cdot 2^3 \cdot 2 = 2 \quad \text{in } \mathbb{F}_3.$$

Together with Section 6, this proves Theorem 4.1.

8 Verification, provenance and historical scope

Theorem 1.1 is an algebraic and real-variable proof for all polynomial degrees; it does not depend on numerical verification. Theorem 4.1 is a complete exact counterexample to a universal assertion, with its finite calculations reproduced below.

8.1 Independent checks

The structural certificate in Appendix A was executed successfully. It was checked against independently implemented calculations with different reductions:

Calculation	modulo 3	modulo 1823
Complementary minor and local polynomial certificate	2	1089
Direct original matrix, split under $j \mapsto -j$	$1 \cdot 2$	$1054 \cdot 449$
Literal 5466×5466 matrix, direct elimination	2	—

In the second row, the two factors are determinants of blocks of size 2733; their products are taken in the indicated fields. The third row uses an integer Jacobi-symbol algorithm and direct Gaussian elimination over $\mathbb{F}_3$, without the Chinese remainder or character reductions. These checks support the certificate. The complete reproducible proof route is the one supplied in the present manuscript. No proof-assistant formalization or external referee assessment is asserted.

8.2 Relation to the supplied corpus

The investigation used the supplied *Verifiable Mathematical Reasoning Corpus*, at commit `c30d40119134a59eda6fa5cc6331f27c8e67d034` [6]. Its work on root criteria, finite congruences, and spectral reductions supplied directions for investigation. Manuscript [077], the T96–T100 collection, led to Sun's determinant problems; its T100 discussion concerns Conjecture 4.10(ii), a different assertion. The finite congruence viewpoint of [089] was another useful heuristic. No unverified corpus theorem is used as a premise here. The targets and their original hypotheses were checked against the primary papers [1, 4].

8.3 What is proved and what is not certified

The mathematical assertions are the original Chasse–Csordas implication and the falsity of Sun's original universal assertion. The stronger local-interval version is stated explicitly rather than substituted for the original target. For the first result, the forward measure formula was already proved in Chasse's dissertation, and the decisive inverse-branch mechanism is already present in Szegő's 1934 argument. This manuscript supplies the full adaptation and cancellation-safe recovery of the polynomial, but makes no claim to have invented that general mechanism.

For the second result, the Chinese remainder theorem, character diagonalization, polynomial Euclidean algorithms and the complementary-minor identity are classical. Li–Yan already use character diagonalization and Chinese remainder lifting in the same determinant family. Here the additional calculations show that the original truncated determinant is a unit at both prime factors of 5469, including the required nonzero complementary inverse minor. This refutes the conjecture; it does not classify all integers in the arithmetic progression, and no minimality claim for 5469 is made. Sun reports tests below 2000 [4, Remark 5.6]; this counterexample lies outside that range.

A bounded literature review checked the exact source conjectures, Chasse's dissertation, the relevant classical rational-measure argument, the dedicated Li–Yan paper, and searches for subsequent resolutions and the specific counterexample. No identical later resolution was located in that review. Such an absence is not a certificate of priority or of the complete state of the literature. The stronger requested success conditions therefore have the following assessment.

Gate	Assessment for the present manuscript
G1	Both targets have prior named and numbered source statements.
G2	The finite-difference quotient, degree, measure law, original determinant, indices, and parameters are retained. Every change of representation is justified.
G3	The source conjectures and inspected later treatments are documented; a universal absence of prior resolutions is not certified.
G4	The polynomial implication is proved for all degrees and all hypotheses in its statement. The determinant assertion is refuted by an exact unconditional counterexample.
G5	The arguments reach actual rigidity and nonvanishing. However, a historically unprecedented general principle is not established; Szegő's prior mechanism is material, and the determinant tools are classical.
G6	No identical resolution was found in the bounded review. Search absence is not proof of firstness, and absolute priority is not asserted.

The manuscript is a mathematical proof document for scrutiny. It does not certify all historical or methodological success conditions, an external referee's acceptance, or a higher significance ranking than every previous manuscript in the corpus.

A Complete exact certificate

The following program needs Python 3.8 or later and only its standard library. It reads no files, downloads no data, and uses no floating-point operations. Polynomial lists are in increasing order

of degree. The displayed code is embedded in this `.tex` file; no separate source file is needed.

The routines `sub`, `mul` and `divrem` perform polynomial arithmetic over the indicated prime field. The extended Euclidean algorithm in `inverse_mod` maintains a coefficient of the input polynomial modulo the modulus. The resultant routine uses, with $R = A \bmod B$, $d = \deg A$, $e = \deg B$, and $r = \deg R$,

$$\operatorname{Res}(A, B) = (-1)^{de} \operatorname{lc}(B)^{d-r} \operatorname{Res}(B, R), \qquad \operatorname{Res}(A, b) = b^{\deg A}.$$

This proves that the computed scalar is the resultant used in (18). Inverse certificates are additionally checked by multiplying them out. Primality of 1823 and 911 is verified by exhaustive trial division through their integer square roots, so no probabilistic primality test is involved.

The output is

```
n = 5469
mod p: Q = 1126 E = 1662 D = 1089
mod 3: det(Ap) = 2 a = 1 b = 0 inverse minor = 2 D = 2
Jacobi(D, 5469) = -1
```

The program uses $Q = \prod_{s=1}^{p-2}(-a_s) = -P$, so its Q^3E is exactly $-P^3E$ from (15).

```
#!/usr/bin/env python3
"""Exact, standard-library-only certificate for Sun Conjecture 5.6(ii).

Every operation is integer arithmetic.  This verifies D={5,5}_5469 is
2 modulo 3 and 1089 modulo 1823, hence (D/5469)=-1, not 0.
The structural formulas are proved in Sections 5--7 of this manuscript.
"""

from math import isqrt


def trim(a):
    while a and not a[-1]:
        a.pop()
    return a


def sub(a, b, ell):
    c = [0] * max(len(a), len(b))
    for i, x in enumerate(a):
        c[i] += x
    for i, x in enumerate(b):
        c[i] -= x
    return trim([x % ell for x in c])


def mul(a, b, ell):
    if not a or not b:
        return []
    c = [0] * (len(a) + len(b) - 1)
    for i, x in enumerate(a):
        if x:
            for j, y in enumerate(b):
                c[i+j] += x*y
    return trim([x % ell for x in c])


def divrem(a, b, ell):
    assert b
    a = a[:]
    q = [0] * max(0, len(a)-len(b)+1)
```

```
    leading_inverse = pow(b[-1], -1, ell)
    while a and len(a) >= len(b):
        k = len(a)-len(b)
        t = a[-1] * leading_inverse % ell
        q[k] = t
        for j, x in enumerate(b):
            a[k+j] = (a[k+j]-t*x) % ell
        trim(a)
    return trim(q), a


def inverse_mod(a, modulus, ell):
    r0, r1 = modulus[:], divrem(a, modulus, ell)[1]
    t0, t1 = [], [1]
    while r1:
        q, r2 = divrem(r0, r1, ell)
        r0, r1 = r1, r2
        t0, t1 = t1, sub(t0, mul(q, t1, ell), ell)
    assert len(r0) == 1 and r0[0], "not invertible"
    t0 = [(x * pow(r0[0], -1, ell)) % ell for x in t0]
    return divrem(t0, modulus, ell)[1]


def resultant(a, b, ell):
    a, b = trim(a[:]), trim(b[:])
    res = 1
    while b:
        m, n = len(a)-1, len(b)-1
        if n == 0:
            return res * pow(b[0], m, ell) % ell
        r = divrem(a, b, ell)[1]
        if not r:
            return 0
        if m*n % 2:
            res = -res
        res = res * pow(b[-1], m-(len(r)-1), ell) % ell
        a, b = b, r
    return 0


def main():
    p = 1823
    n = 3*p
    assert n == 5469 and n % 60 == 9 and n > 69
    assert all(p % d for d in range(2, isqrt(p)+1))
    assert all(911 % d for d in range(2, isqrt(911)+1))
    m = p-1
    g = 5
    # m=2*911, so these two tests certify that g generates F_p^*.
    assert pow(g, m//2, p) != 1 and pow(g, m//911, p) != 1

    def chi(x):
        r = pow(x % p, m//2, p)
        return -1 if r == p-1 else r

    assert chi(5) == -1

    # Modulo p: coefficients of (X^2+5X+5)^((p-1)/2).
    a = [1]
    for _ in range(m//2):
        b = [0]*(len(a)+2)
        for i, x in enumerate(a):
            b[i] += 5*x
```

```
            b[i+1] += 5*x
            b[i+2] += x
        a = [x % p for x in b]
    assert len(a) == p and a[0] == p-1 and a[-1] == 1
    assert all(a[s] for s in range(1, p-1))
    Q = 1
    for s in range(1, p-1):
        Q = Q * (-a[s]) % p
    E = sum(pow(a[s], -1, p) for s in range(2, p-1, 2)) % p
    Dp = pow(Q, 3, p)*E % p
    assert (Q, E, Dp) == (1126, 1662, 1089)
    alpha = sum(pow(a[s], -1, p) for s in range(1, p-1)) % p
    beta = sum((-1)**s * pow(a[s], -1, p)
               for s in range(1, p-1)) % p
    assert (alpha, beta, (-Q) % p) == (921, 580, 697)
    assert pow(-4, p, p)*pow((-Q) % p, 3, p) % p == 641
    assert E*pow(4, -1, p) % p == 1327

    # Modulo 3: multiplicative circulant and its inverse off constants.
    # F_k=chi(g^(2k)+5g^k+5), S=1+X+...+X^(m-1).
    F = []
    z = 1
    for _ in range(m):
        F.append(chi(z*z+5*z+5) % 3)
        z = z*g % p
    assert z == 1 and sum(F) % 3 == 0
    S = [1]*m
    H = inverse_mod(F, S, 3)
    # Independently multiply the returned inverse, checking its certificate.
    assert divrem(mul(F, H, 3), S, 3)[1] == [1]
    H += [0]*(m-len(H))
    kshift = sum(H)*pow(m, -1, 3) % 3
    H = [(x-kshift) % 3 for x in H]
    assert sum(H) % 3 == 0
    cyclic_modulus = [2] + [0]*(m-1) + [1]
    target = [(1-pow(m, -1, 3)) % 3]
    target += [(-pow(m, -1, 3)) % 3]*(m-1)
    assert divrem(mul(F, H, 3), cyclic_modulus, 3)[1] == trim(target)
    aa, bb = H[0], H[m//2]
    det_Ap = m*resultant(S, F, 3) % 3
    u = pow(m, -1, 3)
    det_minor = u*u*(aa+bb)*pow(8, -1, 3) % 3
    D3 = pow(-4, p, 3)*pow(det_Ap, 3, 3)*det_minor % 3
    assert (det_Ap, aa, bb, det_minor, D3) == (2, 1, 0, 2, 2)

    # Legendre symbols at the two distinct prime factors of n.
    assert pow(Dp, m//2, p) == 1 and Dp == 33**2
    assert D3 == 2
    print("n =", n)
    print("mod p: Q =", Q, "E =", E, "D =", Dp)
    print("mod 3: det(Ap) =", det_Ap, "a =", aa, "b =", bb,
          "inverse minor =", det_minor, "D =", D3)
    print("Jacobi(D, 5469) = -1")


if __name__ == "__main__":
    main()
```

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